CODESTOCK 2026

# An Afternoon Project Used by Thousands and How GitHub Actions Maintain It While I Sleep

*Using GitHub Actions to Automate the Boring Stuff*

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Slides, Notes, and Examples

# What are we talking about?

- GitHub Actions
- Sustainable Open Source Maintenance
- Automating the Little Things
- Automating the Weird Things

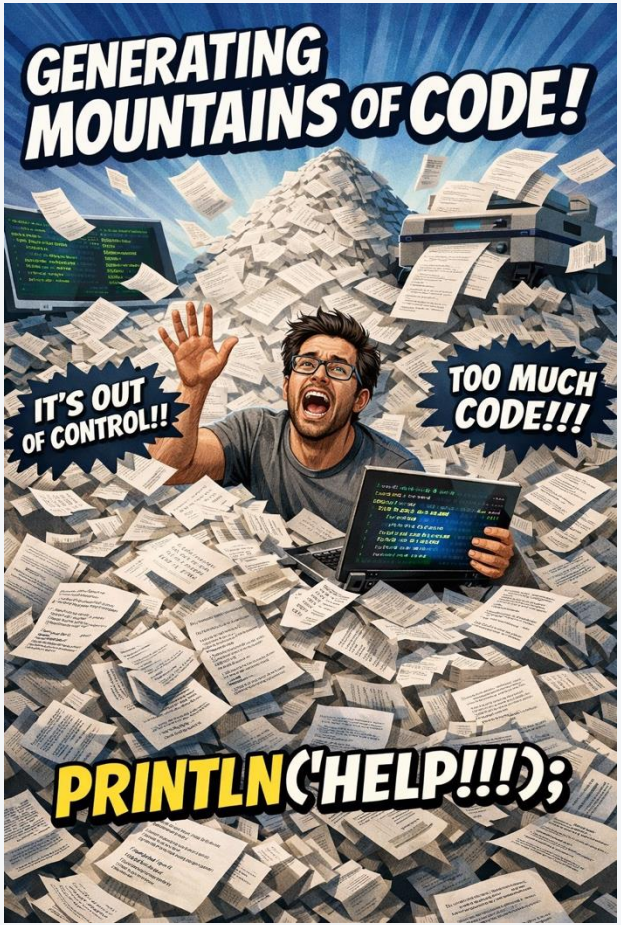
This isn't an AI talk, but....

This isn't an AI talk, but....

We’re Using AI To Generate Mountains of Code

This isn't an AI talk, but....

We’re Using AI To Generate Mountains of Code (and bad memes)



**Kyle Daigle**
@kdaigle

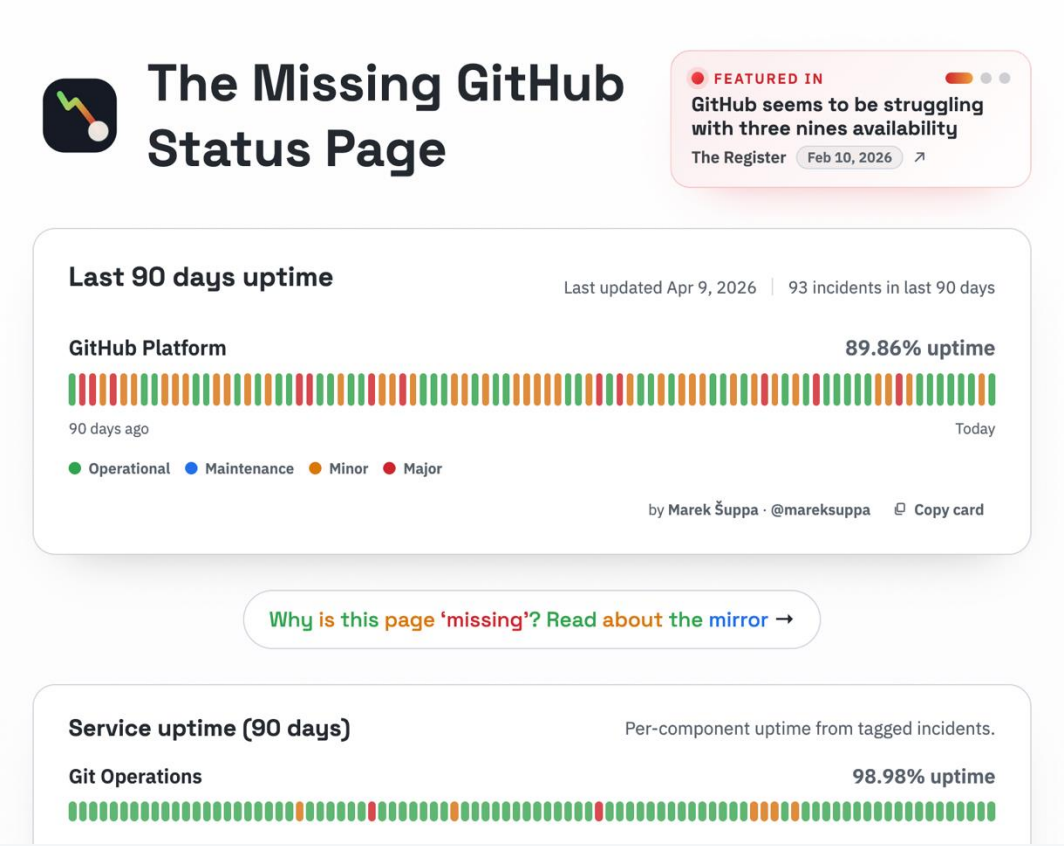
Yup, platform activity is surging. There were 1 billion commits in 2025. Now, it's 275 million per week, on pace for 14 billion this year if growth remains linear (spoiler: it won't.)

GitHub Actions has grown from 500M minutes/week in 2023 to 1B minutes/week in 2025, and now 2.1B minutes so far this week.

So we're pushing incredibly hard on more CPUs, scaling services, and strengthening GitHub's core features.

And as a fine purveyor of hand-crafted shit code for many years, I'm not gonna weigh in on that. 🤣

Kyle Daigle, COO at GitHub




AI has effectively broken the internet. We're seeing it everywhere.

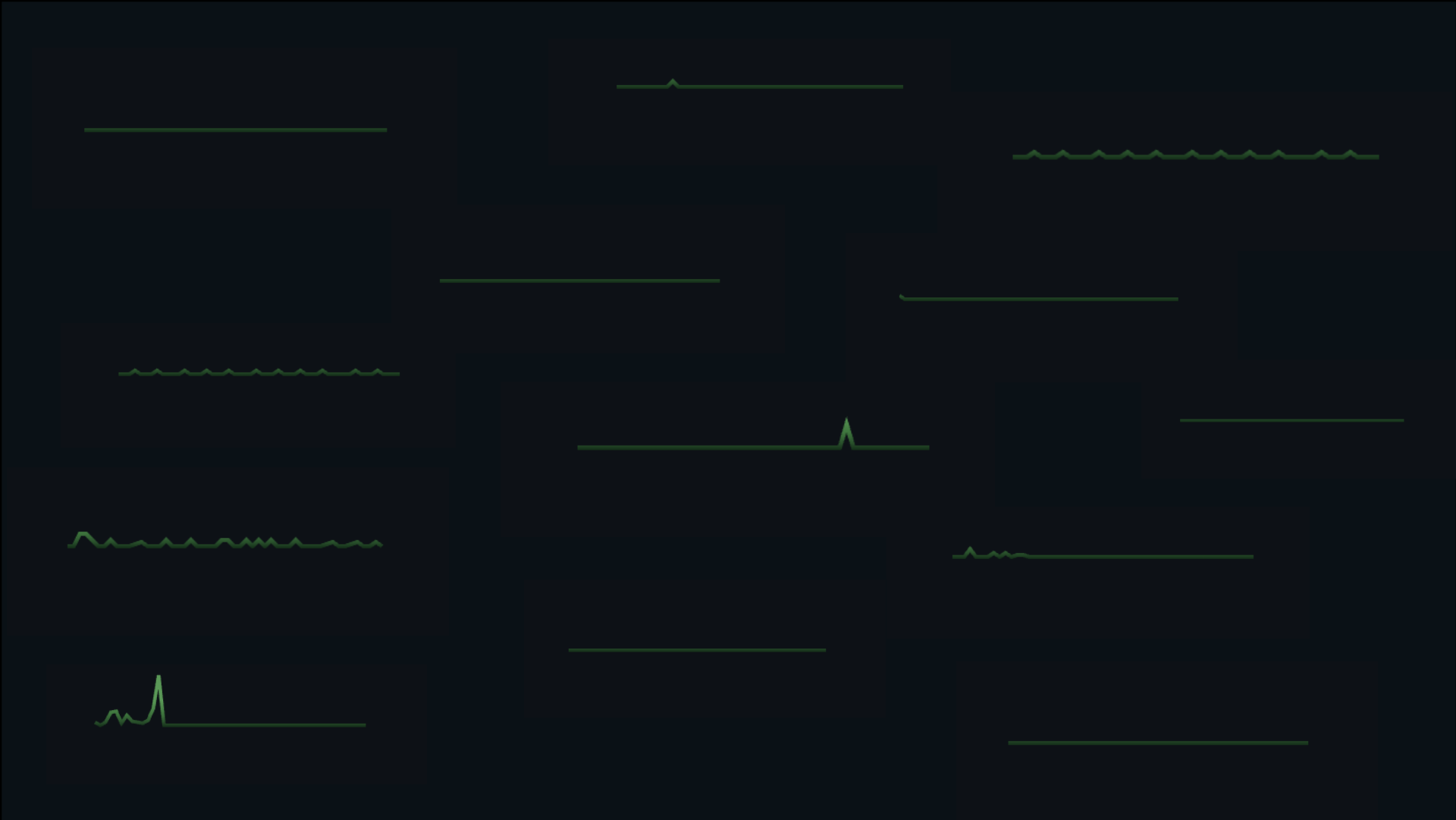
Because we're ~~writing~~ shipping more code than ever,
it is becoming harder and harder to maintain our code.

# This Is An Even Bigger Problem If You Are Easily Distracted...

*Or Lazy? Shiny Object Syndrome?*

“I’m not lazy or easily distracted, I’m efficient!” – My Brain

*points to abandoned projects*



Many of us probably have lots of projects like that. And that’s okay.

# Open Source is Hard

It takes serious dedication to maintain projects over the long term, especially when there's typically minimal financial incentive to do so.

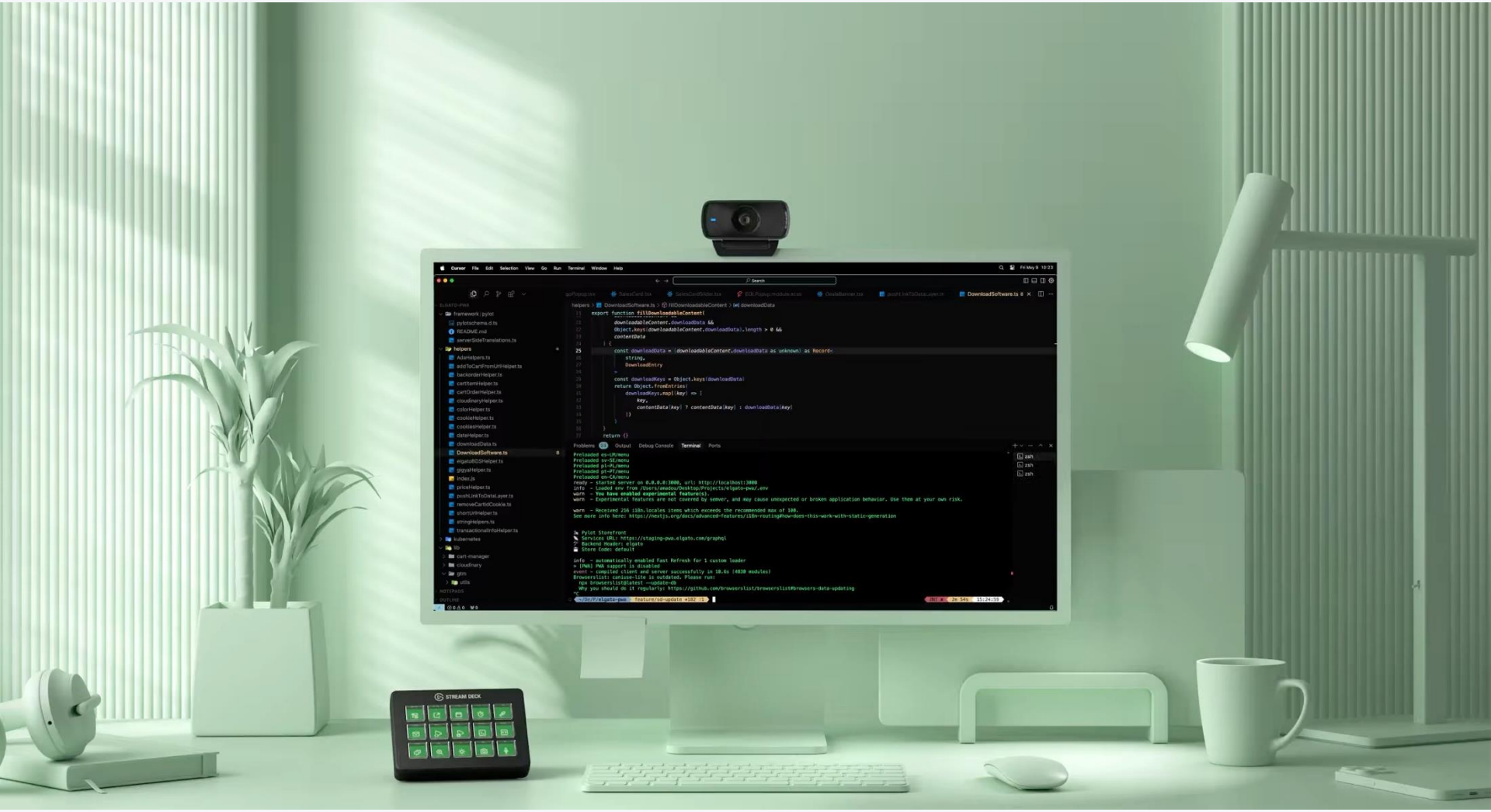
Plus, there are other projects that you will probably be drawn to. Just because you start something, doesn’t mean you have to finish it (fail fast).

With that in mind,
let's travel back in time.

# Welcome to the year 2022

Back in those days we were using primitive tools like, GitHub Copilot’s AI tab autocomplete

And that was then when I bought one of these

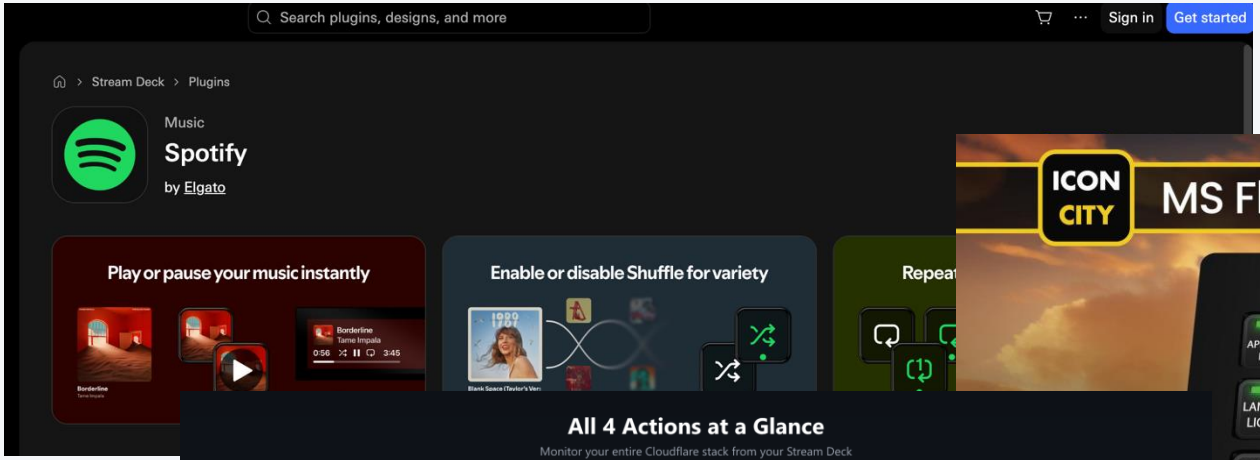

Cursor File Edit Selection View Go Run Terminal Window Help
Fri May 9 10:23
DownloadSoftware.ts
export function fillDownloadableContent(
downloadableContent.downloadData &&
Object.keys(downloadableContent.downloadData).length > 0 &&
contentData
const downloadData = (downloadableContent.downloadData as unknown) as Record<
string,
DownloadEntry
>
const downloadKeys = Object.keys(downloadData)
return Object.fromEntries(
downloadKeys.map(key => [
key,
contentData[key] ? contentData[key] : downloadData[key]
])
)
}
return {}
Problems 11 Output Debug Console Terminal Ports
Preloaded es-LM/menu
Preloaded sv-SE/menu
Preloaded pl-PL/menu
Preloaded pt-PT/menu
Preloaded en-CA/menu
Ready - started server on 0.0.0.0:3000, url: http://localhost:3000
info - Loaded env from /Users/amadou/Desktop/Projects/elgato-pwa/.env
warn - You have enabled experimental feature(s).
warn - Experimental features are not covered by semver, and may cause unexpected or broken application behavior. Use them at your own risk.
warn - Received 216 i18n.locales items which exceeds the recommended max of 100.
See more info here: https://nextjs.org/docs/advanced-features/i18n-routing#how-does-this-work-with-static-generation
Pylot Storefront
Services URL: https://staging-pwa.elgato.com/graphql
Backend Header: elgato
Store Code: default
info - automatically enabled Fast Refresh for 1 custom loader
> [PWA] PWA support is disabled
event - compiled client and server successfully in 10.6s (4820 modules)
Browserslist: caniuse-lite is outdated. Please run:
npx browserslist@latest --update-db
Why you should do it regularly: https://github.com/browserslist/browserslist#browsers-data-updating
STREAM DECK

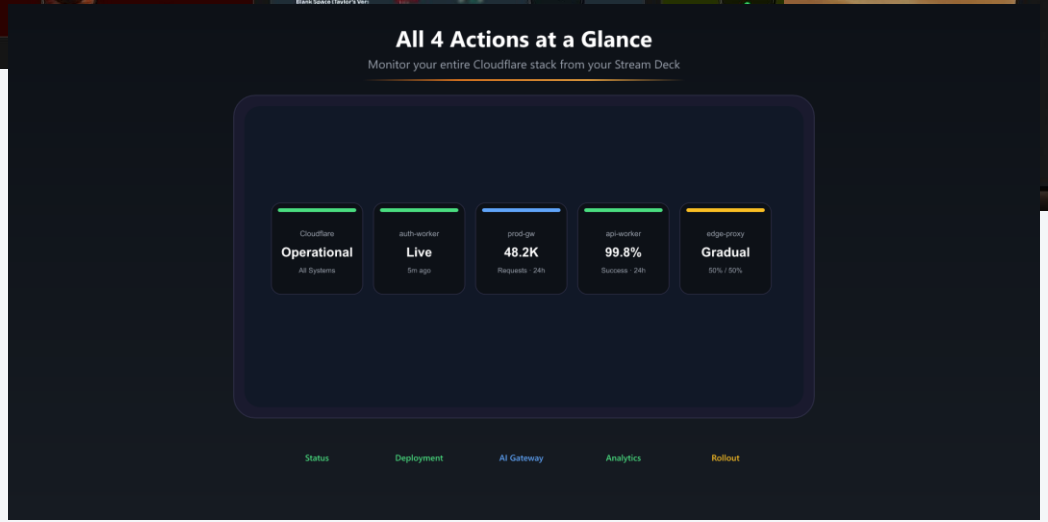
STREAM DECK
I'm Live!

# What Is That?

We've all seen shortcut pads. They're useful for macros, weird characters, etc. But what makes a Stream Deck unique is that the buttons are screens.

This means that you can put any content on the screen that you want.


Search plugins, designs, and more
Sign in
Get started
Stream Deck › Plugins
Music
Spotify
by Elgato
Play or pause your music instantly
Enable or disable Shuffle for variety
Repeat


All 4 Actions at a Glance
Monitor your entire Cloudflare stack from your Stream Deck
Cloudflare
Operational
All Systems
auth-worker
Live
5m ago
prod-gw
48.2K
Requests · 24h
api-worker
99.8%
Success · 24h
edge-proxy
Gradual
50% / 50%
Status
Deployment
AI Gateway
Analytics
Rollout

ICON CITY
MS Flight Simulator 2020
STREAM DECK
APPROACH HOLD
AVIONICS MASTER
CAMERA INSTRUMENTS
ALTITUDE HOLD
LANDING LIGHTS
BEACON
TRAN SPONDER
PUSH BACK
TAXI LIGHTS
REQUEST
GEAR
SPOILERS
PARKING BRAKES
NAV RADIO
PROFILE PACK

# Plugins for everything!

**Volume Controller**
by Elgato · Audio
Ready to take control of your…
Free

**OBS Studio**
by Elgato · Video
Pro production at your…
Free

**Discord**
by Elgato · Social
Talk and hang out with friends…
Free

**Voicemod**
by Voicemod · Audio
Control Voicemod's voice…
Free

**Control Center**
by Elgato · Lighting
Illuminate on command. With…
Free

**Twitch**
by Elgato · Social
Ready to impress your viewer…
Free

**CPU**
by Elgato · Utilities
Monitor your processor in real…
Free

**YouTube**
by Elgato · Social
Stream more, micromanage…
Free

**Audio Switcher**
by Fred Emmott · Audio
Toggle or set the active audio…
Free

**Analog Clock**
by Elgato · Utilities
Turn your Stream Deck into a…
Free

**Wave Link (Legacy)**
by Elgato · Audio
This plugin is compatible with…
Free

**Spotify**
by Elgato · Music
Control Spotify from Stream…
Free

**Clocks**
by Elgato
Never be late again. With the…
Free

**Audio Mute**
by Fred Emmott · Audio
Toggle the mute for audio…
Free

**Streamlabs**
by Streamlabs · Engagement
Control Streamlabs Dashboard.
Free

**Philips Hue (Legacy)**
by Elgato · Lighting
A new and improved version o…
Free

**iCUE Windows**
by Corsair · Utilities
Use your Stream Deck to cont…
Free

**Weather**
by Elgato · Utilities
Stay ahead of the forecast. Th…
Free

**Home Assistant**
by Christoph… · Smart…
Control your Home Assistant…
Free

**Camera Hub**
by Elgato · Video
Got an Elgato camera or…
Free

**Streamlabs Desktop**
by Streamlabs · Streaming

**Govee Controller**
by GeekyEggo · Lighting

**Apple Music**
by Elgato · Music

**Counter**
by Elgato · Utilities

# Manual Process

When I was setting mine up, I found myself going and getting brand icons for opening various websites (LinkedIn, Twitter, etc.). The process was manual:

1. Locate the icon
2. Download the file
3. Edit the file
4. Import into the Stream Deck app.

# Can I Automate This?

I decided there must be a better way, and sure enough, Stream Deck supports custom icon packs so that you can natively select icons eliminating all the in between steps.

So, as any good developer does, I spent a few hours automating a 5-minute problem. 🧠

That's normally a bad idea.
But at scale…it might just be worth it.

# Mostly Automated Process

In an afternoon, I created a Python script to process the files from an icon repository and create an installable icon pack. From there, I submitted the icon pack to the Stream Deck store so that others with the same frustration could easily install icons in one click.

# Shoutout to the Simple Icons team, their work curating the image files make this and other projects possible

simple-icons / simple-icons

Type / to search

Code | Issues 652 | Pull requests 125 | Discussions | Actions | Security and quality 1 | Insights

**simple-icons** Public

Sponsor | Edit Pins | Watch 173 | Fork 3.1k | Starred 24.9k

develop | 2 Branches | 316 Tags | Go to file t | Add file | Code

agj Add Elm package (#14589) ✓ bc857c9 · 48 minutes ago 7,747 Commits

| | | |
|---|---|---|
| .devcontainer | Replace markdown-cli2 by rumdl (#14329) | 2 months ago |
| .github | Fix verify workflow (#14389) | 2 months ago |
| data | Add Fluxer icon (#14477) | 2 days ago |
| icons | Add Fluxer icon (#14477) | 2 days ago |
| scripts | Upgrade xo to 2.0.2 (#14484) | 3 weeks ago |
| tests | Upgrade xo to 2.0.2 (#14484) | 3 weeks ago |
| .dockerignore | Deprecate simple-icons/icons entrypoint (#13530) | 10 months ago |
| .editorconfig | Migrate to prek.toml (#14336) | 2 months ago |
| .gitattributes | Use package lock file (#13269) | 11 months ago |
| .gitignore | Deprecate simple-icons/icons entrypoint (#13530) | 10 months ago |
| .jsonschema.json | Add linting for aliases (#13348) | 11 months ago |
| .node-version | Bump action versions & use Node.js 24 (#13790) | 7 months ago |
| .npmignore | Deprecate simple-icons/icons entrypoint (#13530) | 10 months ago |
| .npmrc | Use package lock file (#13269) | 11 months ago |
| .nvmrc | Bump action versions & use Node.js 24 (#13790) | 7 months ago |
| .prettierignore | Rename _data folder to data (#13316) | 11 months ago |

**About**

SVG icons for popular brands

simpleicons.org

svg, design, design-assets, brand, icons, icon-pack, svg-icons, brand-colors, logo, svg-files, logos, branding, icon-packs, brands, svg-images, brand-assets, iconset, branding-assets, company-brands

Readme
CC0-1.0 license
Code of conduct
Contributing
Security policy
Activity
Custom properties
24.9k stars
173 watching
3.1k forks
Report repository

**Releases** 302

Release 3 new icons and 1... Latest
5 days ago

# It was better, but…

The process took about half an hour to run since it was really just at this point a file processing script.

Still, it worked, so I was happy. I could have stopped there, but I wanted to take it further.

# Why?

- Simple Icons regularly creates new releases. Often this is because companies decided they don't want their logo included and request removal. As a result, updates to the icon pack need to be made quite frequently.

- I don't make any money from this, so it was hard to prioritize. Since I decided to release it to the public, there were expectations and people, maybe some of you, relying on it.

Rather than manually running the script and submitting the icon to Stream Deck's store, I decided to automate it.

# Enter: GitHub Actions 🔥

# TL;DR

- It is nothing new. GitHub Actions were released in 2019.
- Especially today, and somewhat so in 2022 (it doesn't feel like it but Copilot launched in 2021!), AI enables developers like us to quickly ship code.

And what do you know. Actions make a great place to run scripts and automations. It is almost like that’s what they’re designed for.

# What are Actions typically used for?

CI, CD, testing, linting, formatting, builds, deployments, releases, publishing, packaging, versioning, changelogs, SBOMs, pull requests, code review, security scans, dependency updates, vulnerability checks, issue triage, labeling, integration tests, end-to-end tests, static analysis, secret scanning, compliance, infrastructure, …

Automating my project was a perfect use case.

# Fully Automated Process

Using GitHub Actions, I could run the scripts on a cron, go get the latest Simple Icons release, process the image files, create the icon pack file, and then upload it into the Stream Deck store.

Running strong for

# 4

years

Only

# 1

maintainer

Weekly releases requiring an average of

# 0

hours per week of work

Greater than

# 60,000

people have downloaded it

I'm able to always keep the files up to date, generate preview images, and submit the pack to the store all without *any* time commitment.

**The best part?**
**This runs while I'm asleep.**

# Time Savings at Scale

If you're having a problem, others probably are too.

Even if it isn't a sellable product, putting a little effort into something that saves a lot of people time is worth it in my book.

**So, my pitch to you is to automate as much as possible using the tools you already have at your disposal.**

GitHub Actions are free up to a generous number of minutes even in private repos AND there are no logic limitations like those imposed by "automation" tools.

It might not make sense for everything, but with AI, there's nothing stopping you from trying.

# Examples: What can you do with GitHub Actions?

# Level 0 Automate CI/CD:

Most of you are already doing this, but the baseline no matter if your using GitHub Actions, GitLab, Azure Pipelines, whatever is to setup a solid pipeline for your applications.

Run tests, linters, formatters, security scans, code reviews, etc. then automate your deployment process.

It is 2026. This is easy to setup.
Just ask Claude to help create Actions for your application’s stack.

GitHub Actions for Next.js CI/CD pipeline

Share

Create github actions for these things. Assume the project is a Next.js app. Follow best practices and favor free tools. Only respond with the action file(s) within markdown. If there are duplicate items, feel free to merge them into one thing.

CI, CD, testing, linting, formatting, builds, deployments, releases, publishing, packaging, versioning, changelogs, SBOMs, pull requests, code review, security scans, dependency updates, vulnerability checks, issue triage, labeling, integration tests, end-to-end tests, static analysis, secret scanning, compliance, infrastructure

Architected comprehensive GitHub Actions workflows using free tools >

yaml

```yaml
# .github/workflows/ci.yml
name: CI

on:
  push:
    branches: [main, develop]
  pull_request:
    branches: [main, develop]

concurrency:
  group: ${{ github.workflow }}-${{ github.ref }}
  cancel-in-progress: true

jobs:
  lint-and-format:
    name: Lint & Format
```

Reply...

Sonnet 4.6 Extended

Claude is AI and can make mistakes. Please double-check responses.

MJ

# Level 1 Automate The Stuff You Do:

The next step up is automating things like change logs, issue labeling, issue triage, code reviews (with AI), etc.

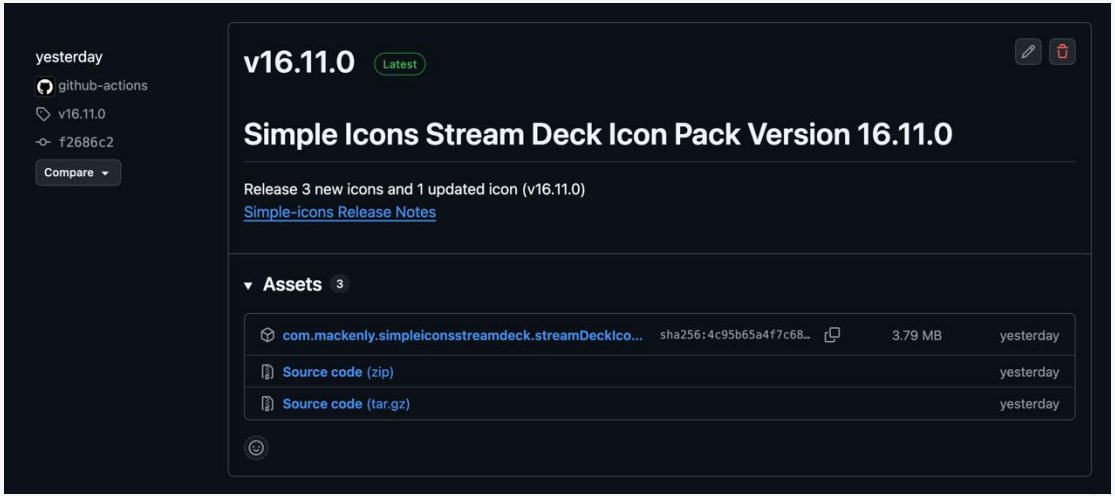
As engineers, our time is extremely valuable. Manually doing things that could be automated keeps us busy but doesn’t let us get anything done.

# Level 2 Use Actions
# As An Automation Platform:

This is where it gets fun.

# Level 2 Use Actions
# As An Automation Platform:

Like me, completely automate the maintenance of your project.



# Level 2 Use Actions
# As An Automation Platform:

Use Actions for Git Scraping

We all know scraping, but when you do it with git you can track changes to the data overtime. This means you get a free time-series database and data scraping platform.

```
149 149     "AcresBurned": 56338,
150 150     "FuelType": null,
151     -   "PercentContained": 90,
    151 +   "PercentContained": 92,
152 152     "ControlStatement": null,
153     -   "ConditionStatement": "<p><b>SHASTA COUNTY DEPARTMENT OF AGRICULTURE/WEIGHTS & MEASURE</p></b>\n\n<p>As
    153 +   "ConditionStatement": "<p><b>SHASTA COUNTY DEPARTMENT OF AGRICULTURE/WEIGHTS & MEASURE</p></b>\n\n<p>As
154 154     "Counties": [
155 155       "Shasta"
156 156     ],
157 157     "CountyIds": "45",
158 158     "SearchDescription": "Zogg Fires started on September 27, 2020 off of Zogg Mine Road and Jenny Bird Lane
159 159     "SearchKeywords": null,
160 160     "AdminUnit": "CALFIRE Shasta-Trinity Unit",
161     -   "Updated": "2020-10-08T06:30:47.7568591-07:00",
    161 +   "Updated": "2020-10-08T17:29:27.785188-07:00",
162 162     "Started": "2020-09-27T16:03:27",
163 163     "Extinguished": null,
164 164     "ArchiveYear": 2020,
    @@ -174,14 +174,14 @@
174 174     "StructuresDamaged": 27,
175 175     "StructuresThreatened": null,
176 176     "StructuresEvacuated": null,
177     -   "PersonnelInvolved": 968,
178     -   "CrewsInvolved": 18,
    177 +   "PersonnelInvolved": 798,
    178 +   "CrewsInvolved": 14,
179 179     "Injuries": 1,
180 180     "Fatalities": 4,
181 181     "Helicopters": 6,
182     -   "Engines": 82,
    182 +   "Engines": 59,
183 183     "Dozers": 14,
184     -   "WaterTenders": 31,
    184 +   "WaterTenders": 27,
185 185     "AirTankers": null
```

https://simonwillison.net/2020/Oct/9/git-scraping/

# ca-fires-history

Public

Watch 1 | Fork 46 | Star 217

main | 1 Branch | 0 Tags | Go to file t | Add file | Code

**Automated** Latest data: Sun Jul 28 22:17:25 UTC 2024 — 9484240 · 2 years ago — 3,705 Commits

| | | |
|---|---|---|
| .github/workflows | Scrape data from new source, refs #5 | 3 years ago |
| README.md | Scrape data from new source, refs #5 | 3 years ago |
| incidents-june-2023-format.json | Latest data: Sun Jul 28 22:17:25 UTC 2024 | 2 years ago |
| incidents.json | Latest data: Sat Mar 4 05:09:26 UTC 2023 | 3 years ago |

## README

# ca-fires-history

Tracking fire data from www.fire.ca.gov/incidents

## As-of June 28th 2023

The scraped data is stored in incidents-june-2023-format.json

## Up to March 2rd 2023

The scraped data was stored in incidents.json

Background on this project: Git scraping: track changes over time by scraping to a Git repository

## About

Tracking fire data from www.fire.ca.gov

fires | disasters | git-scraping

Readme
Activity
217 stars
1 watching
46 forks
Report repository

## Contributors 1

simonw Simon Willison

Code Blame 26 lines (24 loc) · 719 Bytes

Raw

```yaml
name: Scrape latest data

on:
  push:
  workflow_dispatch:
  schedule:
  - cron: '6,26,46 * * * *'

jobs:
  scheduled:
    runs-on: ubuntu-latest
    steps:
    - name: Check out this repo
      uses: actions/checkout@v3
    - name: Fetch latest data
      run: |-
        curl 'https://www.fire.ca.gov/api/sitecore/Incident/GetFiresForMap' \
          -X POST --data-raw 'showFeatured=False' | jq > incidents-june-2023-format.json
    - name: Commit and push if it changed
      run: |-
        git config user.name "Automated"
        git config user.email "actions@users.noreply.github.com"
        git add -A
        timestamp=$(date -u)
        git commit -m "Latest data: ${timestamp}" || exit 0
        git push
```

incidents-june-2023-format.json +3 -3

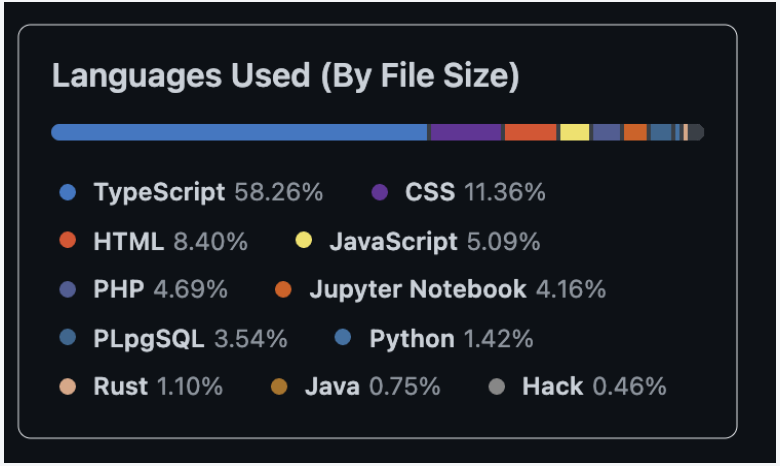
```diff
@@ -97,7 +97,7 @@
 97  97      "County": "Sonoma",
 98  98      "Location": "Stewarts Point Road and Skaggs Springs, Geyserville ",
 99  99      "AcresBurned": 1207,
100      -   "PercentContained": 40,
    100  +   "PercentContained": 50,
101 101      "Longitude": -123.012253,
102 102      "Latitude": 38.705475,
103 103      "Type": null,

@@ -150,8 +150,8 @@
150 150      "AdminUnit": "CAL FIRE Sonoma-Lake-Napa Unit",
151 151      "County": "Colusa",
152 152      "Location": "Sites Lodoga Rd and Wilson Creek, southeast of Stonyford ",
153      -   "AcresBurned": 15564.7,
154      -   "PercentContained": 15,
    153  +   "AcresBurned": 15565,
    154  +   "PercentContained": 5,
155 155      "Longitude": -122.46934,
156 156      "Latitude": 39.31646,
157 157      "Type": null,
```

# Level 2 Use Actions
# As An Automation Platform:

Create Dynamic README Elements

By running an action that gathers data and generates an image file you can easily create dynamic profile elements or display other auto updating data-driven images within your repositories.



mackenly / github-stats

Type / to search

Code | Issues | Pull requests | Agents | Actions | Projects | Wiki | Security and quality | Insights | Settings

Files

master

Go to file t

- .github/workflows
  - main.yml
- generated
  - languages.svg
  - overview.svg
- templates
- .gitattributes
- .gitignore
- LICENSE
- README.md
- generate_images.py
- github_stats.py
- requirements.txt

github-stats / .github / workflows / main.yml

View Runs

mackenly Make it run every other day

f9145df · last year History

Code | Blame | 55 lines (47 loc) · 1.47 KB | Raw

```yaml
name: Generate Stats Images

on:
  push:
    branches: [ master ]
  schedule:
    - cron: "5 0 */2 * *"
  workflow_dispatch:

permissions:
  contents: write

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
    # Check out repository under $GITHUB_WORKSPACE, so the job can access it
    - uses: actions/checkout@v3

```

# Level 2 Use Actions
# As An Automation Platform:

Home Automation

If your office or home office IoT devices have APIs, you can use them in a GitHub Action.

- Turn on a light when a build fails
- Play a fun noise when a deployment succeeds
- Only unlock the door and let the team leave after they've shipped a feature (don't do this)
- Lots of possibilities…

There are truly endless possibilities when you expand your thinking away from GitHub Actions as only something used to process your code.

At the end of the day, you're working with a short-lived virtual machine, that you can use for just about anything. This is more powerful than just a lambda or worker.

# Bonus Tips

# Tip #1: Set timeout-minutes

If something gets hung up, this will prevent you from burning through billable minutes.

I did this once. Don’t make the same mistake as me.

# Tip #2: Use Concurrency

Use concurrency to avoid multiple actions running at the same time. For example, if you merge two changes that will get deployed by CD at nearly the same time, you might have a race condition where the older version ends up overwriting the newest version.

# Thank you.

Enjoy the rest of Code Stock!